Article

Controllable Power Sharing in a Hybrid Fuel-Cell/Battery Vehicle Using Droop Control of Off-the-Shelf Boost Regulators with a Robust Outer-Loop Share Controller

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Abstract: (1) Background: Multi-source DC power systems (hybrid fuel-cell/battery vehicles, DC microgrids) require controllable power sharing between sources, but custom power-electronics solutions are costly; droop control implemented on commercial regulator ICs offers a low-cost alternative if its accuracy and robustness limitations can be managed. (2) Methods: We present a droop-injection architecture in which the feedback nodes of two off-the-shelf boost converters (TI TPS61288) are perturbed by multiplying DACs driven from shunt current sense, yielding an electronically commanded droop resistance; an outer power-share loop is closed by a robust H_∞/Youla-parameterized controller synthesized against a 60-corner uncertain plant family. (3) Results: The synthesized controller achieves $\gamma = 0.686$, exact DC tracking, 11.7 ms delay margin, and worst-corner discrete sensitivity peak of 1.87 versus 26.9 for a nominally tuned PI; an independent 11-state full-order converter model validates the design plant to within 6% in-band over 432 operating points. Hardware bring-up failures (five converter losses) were diagnosed via parasitic-inductance modeling and ideal-diode soft-start timing analysis, and are shown to be generic multi-converter hazards rather than consequences of the droop architecture. (4) Conclusions: Droop injection on commercial ICs plus robust outer-loop synthesis provides accurate, tolerance-immune power sharing without custom power stages.

Keywords: droop control; power sharing; DC microgrid; hybrid vehicle; fuel cell; robust control; H_∞ synthesis; Youla parameterization; boost converter; power path management

1. Introduction

Hybridizing a fuel cell with a battery is the standard remedy for the fuel cell's poor transient response and its intolerance of load steps and reverse current. In such a powertrain both sources feed a common DC link through dedicated converters, and the question of how instantaneous load current is apportioned between them becomes a control problem in its own right. Structurally this is the same problem studied under the heading of DC microgrids, where multiple dissimilar sources share a bus through parallel-connected converters [1–3].

The conventional answer is droop control: each converter deliberately reduces its output voltage in proportion to its output current, so that a virtual output resistance sets the sharing ratio without any communication between units [4,7]. Droop is attractive because it is local, fast and inherently redundant, but it is well documented that it trades bus-voltage regulation against sharing accuracy, and that mismatched line and converter parameters corrupt the achieved split [5,6]. A large body of work therefore adds a secondary or distributed layer that trims the droop coefficients or the voltage references [8–10].

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Almost without exception, that literature presumes a converter whose inner loop the designer owns: modulation is generated in a DSP or FPGA, and the droop term is simply added to a software voltage reference. Practitioners building small hybrid systems rarely enjoy that freedom. Power stages are increasingly bought as monolithic, internally-compensated regulator ICs, in which the control loop is sealed and the only user-accessible node is the feedback divider. For such parts the standard droop recipe has no point of entry, and designers fall back on fixed resistive droop, on a hard master–slave split, or on abandoning controllable sharing altogether.

This paper shows that controllable droop can nevertheless be obtained from such a device. A multiplying digital-to-analogue converter (MDAC) injects a current-proportional perturbation into the feedback node of each off-the-shelf boost regulator, synthesizing a commanded, continuously variable output resistance in analogue hardware. The fast sharing action remains local to each converter; a slow digital outer loop merely commands the droop ratio, and hence the steady-state power split, between the fuel-cell and battery branches.

Because the resulting plant is assembled from an IC whose internal compensation is only approximately published, a sense amplifier and divider network with finite tolerance, and an operating point that moves with bus voltage and load, its parameters are known only within a box rather than exactly. This is precisely the setting for robust synthesis, and $\mathcal{H}_\infty$ and related methods have already been applied to microgrid voltage and sharing loops [22–24]. We accordingly design the outer share controller by $\mathcal{H}_\infty$/Youla synthesis over the parameter corners rather than by tuning a PI at nominal.

The contributions of this work are: (i) a feedback-injection droop mechanism that gives commanded power sharing from internally-compensated, off-the-shelf boost regulators; (ii) a small-signal model of the two-converter droop-sharing loop that accounts for the injection attenuation and the current-sense chain, validated against an independently constructed full-order model; (iii) a robust outer share controller with a documented synthesis and verification procedure; and (iv) experimental validation on a scale fuel-cell/battery vehicle.

The remainder of the paper is organized as follows. Section 2 reviews power-sharing methods and positions the proposed approach. Section 3 develops the droop design and gain selection; Section 4 the robust share controller; Section 5 the board and power-path hardware; Section 6 the bring-up diagnosis; Section 7 the MIMO outlook.

2. Related Work

2.1. Droop Control and Its Refinements

Conventional voltage–current droop emulates a series resistance R_d at each converter output, so that $v_o = V^* - R_d i_o$. Sharing accuracy improves with larger R_d while bus regulation degrades, a tension that is the starting point of most subsequent work [1,3]. Virtual-resistance implementations move this behaviour into the controller rather than dissipating it, and adaptive schemes vary R_d online according to converter rating, state of charge or measured sharing error [6,7,11].

Because purely local droop cannot remove the steady-state voltage deviation, a secondary layer is usually added. Distributed and consensus-based schemes exchange voltage and current estimates over a sparse communication graph to restore the bus and equalize per-unit currents simultaneously [5,8,10], and plug-and-play designs provide stability guarantees that are independent of the network topology [9]. These methods achieve excellent accuracy at the cost of a communication link and its associated latency and failure modes.

2.2. Alternatives to Droop

Active current-sharing methods predate the microgrid framing. Master–slave and average-current-sharing architectures for paralleled supply modules were classified early and remain the industrial default where the modules are identical and co-located [12,13]. They give near-exact sharing but require a dedicated share bus and degrade poorly when the master fails.

DC bus signaling dispenses with an explicit link by encoding mode transitions in the bus voltage itself, so that each unit infers the system state locally [14,15]. The method is elegant for coarse mode arbitration but does not deliver a continuously commandable split.

Optimization-based supervision, particularly model predictive control, has been applied extensively to fuel-cell hybrid powertrains to schedule the source split against fuel economy and degradation objectives [16–18]. Such strategies operate above the converter loops: they decide *what* the split should be, and still require an actuation mechanism to enforce it. The present work is complementary rather than competing, and in fact supplies exactly that mechanism.

2.3. Power Sharing in Fuel-Cell Hybrid Vehicles

In fuel-cell hybrids the split is not arbitrary: the stack must be protected from fast transients, which are absorbed by the battery or a supercapacitor, while the stack carries the slow-moving mean demand [19–21]. This frequency separation is usually enforced by filtering the demand signal or by a rule-based supervisor. Droop achieves the same separation as a structural property when the branches are given different output impedances, which is one reason droop-based energy management has been reported for fuel-cell tramways and hybrid buses [20,21].

2.4. Robust Control of Sharing Loops

Robust synthesis has been used to certify microgrid loops against parameter drift, constant-power-load negative impedance and communication delay [22–24]. These results confirm the value of a worst-case formulation, but they are invariably applied to converters whose internal dynamics are designed alongside the controller. Applying the same discipline to a loop whose actuator is an MDAC perturbing the feedback pin of a commercial regulator, and whose plant gain therefore inherits vendor tolerances, appears not to have been reported.

2.5. Positioning

Table 1 summarizes the trade-offs. The proposed approach occupies a cell that the existing methods do not: it retains the commandability of a centralized scheme and the robustness of a droop scheme, while requiring neither a custom power stage nor a communication link inside the fast loop.

Table 1. Comparison of DC power-sharing methods against the criteria relevant to a low-cost hybrid powertrain. Ratings are qualitative: ••• = favourable, • = unfavourable.

Method	Hardware cost	Comms required	Split controllability	Robustness to converter tol.	Off-the-shelf ICs usable	Refs
Fixed resistive droop	•••	none	•	•	•••	[1,3]
Virtual-resistance droop	••	none	••	••	•	[6,7]
Adaptive / consensus droop	••	sparse	•••	•••	•	[5,8,10]
Master–slave current share	•	share bus	•••	••	•	[12]
Average current sharing	•	share bus	••	••	•	[12,13]
DC bus signaling	•••	none	•	••	••	[14,15]
Supervisory MPC / EMS	••	to supervisor	•••	n/a [a]	n/a [a]	[16–18]
This work: MDAC feedback-injection droop	•••	none [b]	•••	•••[c]	•••	—

[a] Supervisory strategies presuppose an underlying actuation mechanism and are therefore complementary to, not substitutes for, the rows above. [b] No communication inside the fast sharing loop; a low-rate link carries only the share setpoint. [c] By construction of the robust outer controller over the parameter corners; see Section 4.

3. Droop Control Design and Gain Selection

3.1. Design Ethos: Power Sharing Without a Custom Power Stage

The power-sharing objective for a hybrid fuel-cell/battery vehicle is to place a commanded fraction α_{req} of the bus current on the fuel cell and the remainder on the battery, continuously and under load transients. The conventional route is a purpose-designed dual-input converter with a shared, digitally arbitrated current loop. This work deliberately takes the opposite route: both sources are boosted by *unmodified, off-the-shelf* converters (Texas Instruments TPS61288, constant-off-time peak-current-mode boost), and sharing is imposed by *perturbing each regulator's feedback node* with a signal proportional to that regulator's own output current. Each converter therefore behaves as a Thévenin source with an electronically programmable output resistance, and the two sources share current by classical droop — but with a droop coefficient the microcontroller can rewrite at any instant.

The advantages that motivate this choice are practical rather than theoretical:

1. **No custom power stage.** The converters keep their datasheet compensation, protection, soft-start, and thermal behaviour. Nothing in the high-current path is bespoke, so the failure modes are the vendor's characterised ones.
2. **The fast loop stays analog.** Current sensing, scaling, and injection are continuous-time hardware; the digital controller only *trims a coefficient*. Sharing does not collapse if the MCU is late, and the loop is not sampled at the converter's switching time scale.
3. **Graceful degradation.** With the digital-to-analog converters parked at a fixed code the board is still a working, statically drooped parallel supply. The control law is an enhancement, not a prerequisite for operation.

The per-channel signal chain that realises this is fixed by the board and is shown in Figure 1: the channel's output current is measured by an INA253A1 integrated shunt amplifier, scaled by a 12-bit AD5443 multiplying DAC (MDAC) operating in voltage-switching mode, buffered and amplified by an OPA197, and injected through a resistor into the boost converter's feedback divider node.

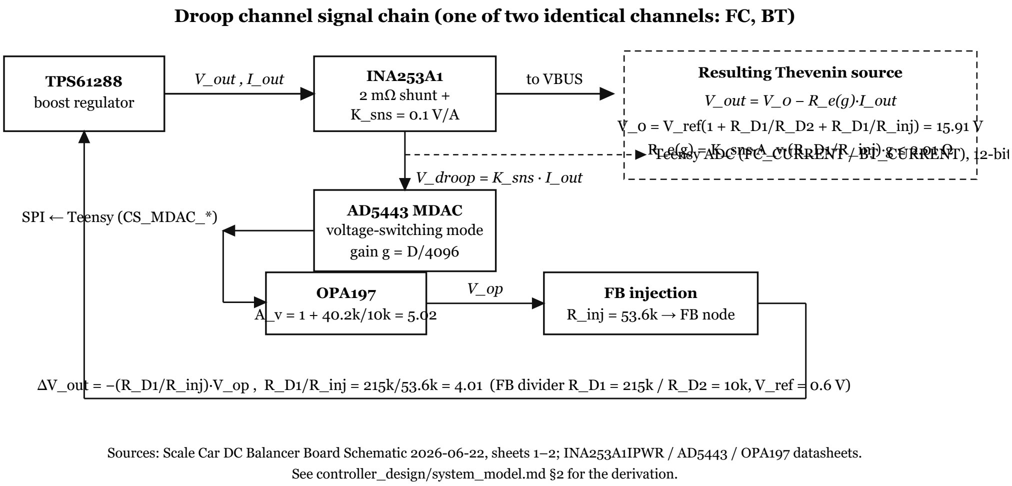

Figure 1. Per-channel droop signal chain. The channel output current is sensed by the INA253A1 ($K_{sns} = 0.1\,\mathrm{V/A}$), scaled by the AD5443 MDAC (gain $g \in [0, 4095/4096]$, written over SPI), amplified by the OPA197 ($A_v = 5.02$), and injected through $R_{inj} = 53.6\,\mathrm{k\Omega}$ into the TPS61288 feedback node ($R_{D1} = 215\,\mathrm{k\Omega}$, $R_{D2} = 10\,\mathrm{k\Omega}$). The converter is unmodified; the entire droop mechanism is an additive perturbation of its feedback divider.

3.2. Droop Equations and the Actuator Mapping

3.2.1. Effective output resistance

The converter servos its feedback node to V_{ref}. Writing KCL at that node (feedback pin current negligible) with the injected voltage V_{op} gives

$$\frac{V_{out} - V_{ref}}{R_{D1}} + \frac{V_{op} - V_{ref}}{R_{inj}} = \frac{V_{ref}}{R_{D2}}, \tag{1}$$

which rearranges into an explicit Thévenin form,

$$V_{out} = \underbrace{V_{ref}\left(1 + \frac{R_{D1}}{R_{D2}} + \frac{R_{D1}}{R_{inj}}\right)}_{V_0 \text{ (no-load setpoint)}} - \frac{R_{D1}}{R_{inj}}\,V_{op}. \tag{2}$$

Substituting the measured-current path $V_{op} = A_v\,g\,K_{sns}\,I_{out}$ yields the programmable output resistance that is the heart of the scheme:

$$V_{out} = V_0 - R_e(g)\,I_{out}, \qquad \boxed{R_e(g) = K_{sns}\,A_v\,\frac{R_{D1}}{R_{inj}}\,g \;=\; R_{e,max}\,g} \tag{3}$$

with the numerical values of the as-built board:

$$K_{sns} = 0.1\ \mathrm{V/A} \qquad \text{INA253A1 fitted (A3 would be 0.4)} \tag{4}$$

$$A_v = 1 + \tfrac{40.2\,\mathrm{k}}{10\,\mathrm{k}} = 5.02 \qquad \text{op-amp gain} \tag{5}$$

$$R_{D1}/R_{inj} = \tfrac{215\,\mathrm{k}}{53.6\,\mathrm{k}} = 4.011 \qquad \text{injection attenuation} \tag{6}$$

$$R_{e,max} = 0.1 \times 5.02 \times 4.011 = \mathbf{2.014\ \Omega} \tag{7}$$

$$V_0 = 0.6 \times 26.51 = 15.91\ \mathrm{V} \qquad \text{no-load setpoint} \tag{8}$$

The 12-bit MDAC resolves this into $\Delta R_e = R_{e,max}/4096 = 0.49\,\mathrm{m\Omega}$ per LSB, which is far below any effect the current sensing can resolve; MDAC quantisation is therefore omitted from the linear model.

3.2.2. Two sources in parallel

With both channels conducting onto the bus through their ideal-diode switches, and total load current I_{tot},

$$V_{bus} = V_{0F} - R_F I_F = V_{0B} - R_B I_B, \qquad I_F + I_B = I_{tot}. \tag{9}$$

The design choice that makes this a well-conditioned control problem is to command the two droop resistances *reciprocally* from a single scalar ratio r, using a fixed design droop scale k_d (denoted `K_DROOP` in firmware):

$$R_F = \frac{k_d}{r}, \qquad R_B = \frac{k_d}{1-r}. \tag{10}$$

This has the property $R_F \parallel R_B = k_d$ *independently of* r: the bus's aggregate droop stiffness is constant while the split is swept. The resulting share is

$$\boxed{\alpha = \frac{I_F}{I_{tot}} = r + \underbrace{\frac{\Delta V_0\, r(1-r)}{k_d\, I_{tot}}}_{d \text{ (output disturbance)}}} \tag{11}$$

where $\Delta V_0 \equiv V_{0F} - V_{0B}$ is the channel-to-channel no-load setpoint mismatch. Three consequences drive the controller design: the nominal static gain is exactly unity ($\alpha = r$); setpoint mismatch enters as an additive output disturbance that is worst at $r = 0.5$ and at light load and scales as $1/k_d$; and the same mismatch perturbs the plant gain, $\partial\alpha/\partial r = 1 + \Delta V_0(1-2r)/(k_d I_{tot})$, which is carried into synthesis as parametric gain uncertainty.

3.2.3. The actuator mapping — and a design lesson

Combining (3) and (10) gives the mapping the firmware must implement:

$$\boxed{g_F = \frac{k_d}{R_{e,max}\, r}, \qquad g_B = \frac{k_d}{R_{e,max}\,(1-r)}} \tag{12}$$

It is worth reporting that the original firmware did *not* implement (12). It used an equivalent-resistance constant $k_{eq} = 0.45\,\Omega$ and computed $g = k_{eq}/(r\, K_{sns} A_v)$ — an expression that models the sense-and-amplify path but silently *omits the feedback-injection attenuation* R_{D1}/R_{inj}, at that time $237\,\text{k}/53.6\,\text{k} = 4.42$. The consequence is not a mild gain error. The commanded gain becomes $g_F = 0.45/(0.502\, r) = 0.896/r$, which exceeds unity for every $r < 0.896$; the SPI write routine clamps to full scale; and therefore across essentially the whole commanded range *both* MDACs sat pinned at $g = 1$, giving $R_F = R_B = R_{e,max}$ and a fixed $50/50$ share.

The control-theoretic reading is the important one: the loop was not badly tuned, it was operating on a **zero-gain plant**. Any $\partial\alpha/\partial r$ was identically zero over the operating range, so no choice of controller gains — and the legacy design used $K_p = K_i = 1$, which (11) makes superficially plausible since the nominal plant gain is unity — could have produced closed-loop share authority. The defect is invisible to a firmware unit test (the code computes what it says it computes), invisible to a step response at fixed load (the share is stable, merely uncontrolled), and only becomes obvious when the actuator mapping is re-derived from the schematic and the saturation limit $g \leq 1$ is checked explicitly. The lesson we draw, and recommend, is that *an actuator-range audit against the physical hardware constants should be a mandatory step before any plant identification*: a saturated actuator and a well-behaved plant are indistinguishable in closed-loop data.

3.3. Gain Selection and the PCB Constraints It Imposed

The droop gain choice is not a purely digital decision. Because R_e is realised in analog hardware, selecting k_d simultaneously fixes MDAC range utilisation, op-amp headroom, and the achievable share authority — and those in turn dictated component values on the PCB. The binding constraints are the following.

Constraint 1: MDAC saturation sets the ratio span.

Since $g \leq 1$, (12) bounds the usable ratio range by $k_d \leq R_{e,max} \cdot \min(r_{min}, 1 - r_{max})$. This is a direct trade of *droop stiffness against control authority*:

Table 2. Droop-scale bound versus commanded ratio span, at $R_{e,max} = 2.014\ \Omega$. The middle row is the shipped design point.

Ratio span $[r_{min}, r_{max}]$	max k_d (Ω)	Character
[0.10, 0.90]	0.201	Wide authority, weak droop
[**0.15**, **0.85**]	**0.302**	**Shipped:** $k_d = 0.30\ \Omega$**, max** $g = 0.993$
[0.20, 0.80]	0.403	Strong droop, narrow authority

The selected point is $k_d = 0.30\ \Omega$ with clamps $[\texttt{DROOP\_R\_MIN}, \texttt{DROOP\_R\_MAX}] = [0.15, 0.85]$, leaving 0.7% of MDAC range as margin against component tolerance. Larger k_d suppresses the mismatch disturbance $d \propto 1/k_d$ but costs bus regulation — the droop sag at $r = 0.5$ is $k_d I_{tot}$, i.e. 1.8 V at 6 A — and narrows the reachable share range. Smaller k_d widens authority but lets the ΔV_0 mismatch dominate the share error.

Constraint 2: the injection resistor sets $R_{e,max}$ *and* the bus setpoint simultaneously.

R_{inj} appears in both (2) and (3): it attenuates the injected droop signal (setting the ceiling $R_{e,max}$) and it also loads the feedback divider (shifting the no-load setpoint V_0). It cannot be chosen for droop range alone. The board's $R_{inj} = 53.6\,\mathrm{k}\Omega$ against $R_{D1} = 215\,\mathrm{k}\Omega$ yields $R_{e,max} = 2.014\ \Omega$ — comfortably above the $0.302\ \Omega$ requirement, i.e. the design deliberately uses only $\approx 15\%$ of the available droop authority at the range centre so that the whole span stays clear of MDAC saturation. This coupling was also the mechanism by which a later system-level decision propagated back into the droop design: the 16 V bus retune (undertaken for converter overshoot headroom, unrelated to sharing) changed R_{D1} from 237 k to 215 kΩ, which moved $R_{e,max}$ from $2.220\ \Omega$ to $2.014\ \Omega$ and required k_d to be reduced from 0.33 to $0.30\ \Omega$ to preserve the $g \leq 1$ bound.

Constraint 3: sense gain versus op-amp headroom.

The OPA197 is supplied from the board's 5 V rail (a post-manufacturing change from 3.3 V), so its output ceiling is ≈ 4.9 V. The injected signal is $V_{op} = A_v g K_{sns} I_{out}$, so the headroom limit is $g\, I_{out} \leq 4.9/(5.02 \times 0.1) = 9.8$ A — non-binding for this vehicle, whose per-channel current stays below roughly 3 A. Note, however, that this margin is a *consequence of a component substitution error*: the design intended the INA253A3 ($K_{sns} = 0.4$ V/A) and the INA253A1 (0.1 V/A) was fitted. The A3 part would have raised $R_{e,max}$ fourfold (making the MDAC range constraint far looser) but would equally have reduced the headroom ceiling to 2.4 A·g, which *would* have been binding at peak current. The firmware carries K_{sns} as an explicit named constant precisely so this variant choice remains a single-line change if the board is respun.

Constraint 4: the clamps are the anti-windup boundary.

Because the ratio clamps $[0.15, 0.85]$ are the true actuator saturation, they — not an arbitrary controller output limit — define the anti-windup boundary for the outer share controller. The clamp is applied in the firmware before the mapping (12) is evaluated, so the commanded MDAC gain is guaranteed to satisfy $g \leq 1$ by construction and the digital controller never operates against a hidden hardware limit. The legacy [0.01, 0.99] clamps, which permitted the saturated regime of Section 3.2.3, were removed from the droop path.

Figure 2 places these elements in the linearised design loop used for synthesis, showing where the mismatch disturbance d and the current-sense noise n enter.

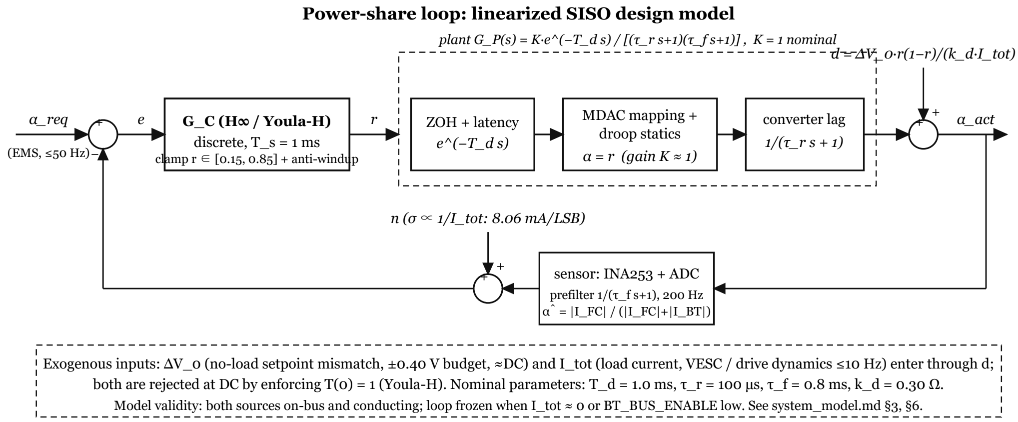

Figure 2. Linearised SISO share-control loop. The controller commands the droop ratio r; the actuator mapping $g = k_d/(R_{e,max} r)$ and its clamp to $[0.15, 0.85]$ form the saturation element. The plant reduces to a near-unity static gain in series with the converter lag τ_r, the measurement prefilter τ_f, and the dominant loop delay T_d. Setpoint mismatch ΔV_0 enters as the additive output disturbance $d = \Delta V_0 r(1-r)/(k_d I_{tot})$; current-sense quantisation enters as output noise n with variance scaling as $1/I_{tot}$.

4. Robust Power-Share Controller Design

The energy-management supervisor running on the vehicle's companion computer emits a requested power-share $\alpha_{req} \in [0, 1]$ at up to 50 Hz. Realising that request on hardware is the job of an inner loop that adjusts the analogue droop resistance of each of the two boost converters until the measured current split matches the request. This section derives the plant seen by that loop, argues why a nominal-tuned PI controller is inadequate for it, synthesises a robust replacement by $\mathcal{H}_\infty$ mixed-sensitivity design followed by a Youla parameterisation gain adjustment, and validates the result against both a 60-corner uncertainty family and an independently constructed full-order converter model.

4.1. Plant Model of the Droop-Controlled Share Loop

4.1.1. Per-channel programmable output resistance

Each source channel (fuel cell, FC; battery, BT) is a synchronous boost converter whose output voltage is drooped in analogue hardware in proportion to its own output current. The sensed current passes through a current-sense amplifier of gain $K_{sns} = 0.1\,\mathrm{V\,A^{-1}}$, a 12-bit multiplying DAC of programmable gain $g \in [0, 1]$, and a non-inverting stage of gain $A_v = 1 + R_{op2}/R_{op1} = 5.02$, and is then injected into the regulator feedback node through $R_{inj} = 53.6\,\mathrm{k\Omega}$ against the divider $R_{D1} = 215\,\mathrm{k\Omega}$, $R_{D2} = 10\,\mathrm{k\Omega}$. Applying Kirchhoff's current law at the feedback node, which the regulator servos to V_{ref}, gives a Thévenin source with an *electronically programmable* output resistance

$$V_{out} = V_0 - R_e(g)\, I_{out}, \qquad R_e(g) = K_{sns} A_v \frac{R_{D1}}{R_{inj}}\, g \;=\; R_{e,max}\, g. \tag{13}$$

With the fitted values, $R_{D1}/R_{inj} = 4.0112$, $V_0 = 15.91\,\mathrm{V}$ and $R_{e,max} = 2.014\,\Omega$, giving a droop resolution of $0.492\,\mathrm{m\Omega}$ per DAC code.

The factor R_{D1}/R_{inj} — the attenuation of the injection path by the feedback divider — was omitted in the original firmware droop mapping, which therefore commanded DAC gains $g > 1$ over almost the entire ratio range and left both DACs clamped at full scale. The share loop consequently saw a *zero-gain* plant, a defect invisible to any nominal-model simulation and only exposed by writing (13) out against the schematic (Section 3.2.3).

4.1.2. Two channels in parallel: the static share law

Both Thévenin sources feed the DC bus through ideal-diode switches. With $V_{bus} = V_{0F} - R_F I_F = V_{0B} - R_B I_B$ and $I_F + I_B = I_{tot}$, and adopting the droop-ratio mapping $R_F = k_d/r$, $R_B = k_d/(1-r)$ — for which the parallel droop $R_F \parallel R_B = k_d$ is independent of r — the measured share obeys

$$\alpha \;\equiv\; \frac{I_F}{I_{tot}} \;=\; r \;+\; \underbrace{\frac{\Delta V_0\, r(1-r)}{k_d\, I_{tot}}}_{d}, \qquad \Delta V_0 \equiv V_{0F} - V_{0B}. \tag{14}$$

Three structural facts follow, and they set the entire control problem.

1. **The nominal static gain is exactly unity** ($\alpha = r$ when $\Delta V_0 = 0$), by construction of the k_d/r mapping. This is precisely why a PI controller with $K_p = K_i = 1$ appeared adequate on paper.
2. **Set-point mismatch enters as an additive output disturbance** d, worst at $r = 0.5$ and at light load, and inversely proportional to the droop scale k_d. With a plausible per-board $\Delta V_0 = 0.4\,\mathrm{V}$ and $k_d = 0.30\,\Omega$, $d = 0.33/I_{tot}$, i.e. **0.17 share at** 2 A. This is a very large static offset and is the reason exact integral action is a hard requirement rather than a refinement.
3. **The same mismatch perturbs the plant gain**, $\partial\alpha/\partial r = 1 + \Delta V_0(1-2r)/(k_d I_{tot})$, which is treated as parametric gain uncertainty $K \in [1-\delta, 1+\delta]$.

The actuator constraint $g \le 1$ bounds the usable authority: $k_d \le R_{e,max} \min(r_{min}, 1 - r_{max})$. The design adopts $r \in [0.15, 0.85]$ with $k_d = 0.30\,\Omega$ (hard bound $0.3020\,\Omega$), which trades a 1.8 V bus sag at 6 A against a manageable mismatch disturbance. Twelve-bit DAC quantisation moves the share by $\approx 3.7 \times 10^{-5}$ per code at the range edge and is negligible against the sensor noise floor.

4.1.3. Dynamics and the design plant

The share statics are, remarkably, *instantaneous*: with $\Delta V_0 = 0$ both channel currents carry the common factor $(V_0 - V_{bus}(t))$, so $\alpha(t) = G_F/(G_F + G_B) = r$ at every instant regardless of the bus-voltage transient. The bus-capacitance pole therefore appears in the share response only through the mismatch term, with residue $\propto \Delta V_0/(k_d I_{tot})$, and is absorbed into the uncertainty description rather than modelled explicitly. What remains is the digital layer plus the converter voltage loops, giving the design plant

$$G_P(s) \;=\; \frac{K\, e^{-T_d s}}{(\tau_r s + 1)(\tau_f s + 1)}. \tag{15}$$

Table 3. Design-plant parameters and uncertainty ranges.

Parameter	Nominal	Range	Physical origin
K	1	$[0.55,\ 1.45]$	static mismatch gain term, Eq. (14)
T_d	1.0 ms	$[0.5,\ 2.0]$ ms	zero-order hold + compute-to-apply latency at $T_s = 1$ m
τ_r	100 µs	$[20,\ 300]$ µs	converter voltage-loop lag, lumped with sense and bus p
τ_f	0.8 ms	$\{0,\ 0.8\}$ ms	optional 200 Hz measurement prefilter

The loop delay budget is $T_d = T_s/2 + T_s + \varepsilon \approx 1.5$ ms worst case at the chosen 1 kHz rate, which gives $\geq 20\times$ oversampling of the intended 5 Hz to 10 Hz closed-loop bandwidth while leaving the fast firmware tick free for fault detection and motor control.

The plant is thus *delay-dominated with near-unity gain*. The lumping of the two converter voltage loops into the single lag τ_r is justified by time-scale separation of about 2.5 decades: the converter loops cross at 6 kHz to 22 kHz (from datasheet compensator arithmetic with $R_C = 61.2\,\mathrm{k\Omega}$ on *both* channels), whereas the share loop crosses near $110\,\mathrm{rad\,s^{-1}}$. Even the slowest modelled $\tau_r = 300\,\mu\mathrm{s}$ contributes only 1.9° of phase at the share crossover. Section 4.4.2 replaces this argument with empirical evidence.

4.2. Why Robustness, and Not Tuning, Is the Issue

Equation (14) makes the plant look trivial: unity DC gain, one small lag, one small delay. That appearance is exactly the trap. The unity gain is a property of the *nominal* hardware, and the two channels are not nominally identical in practice:

- **Regulator tolerance bands are wide.** The converter feedback reference is specified over 0.588 V to 0.612 V across process and temperature (± 2 %), and with ± 1 % divider resistors the channel-to-channel no-load voltage difference budgets to ± 3.4 %, i.e. $\Delta V_0 \approx \pm 0.6$ V worst case. Through Eq. (14) this is simultaneously a large output disturbance *and* a plant-gain perturbation.
- **Ceramic output capacitance derates strongly with bias.** The three 22 µF output capacitors present 66 µF nominally but ≈ 30 µF derated, which moves each converter's voltage-loop crossover by better than a factor of two and hence moves τ_r.
- **The right-half-plane zero moves with load and duty.** f_{RHPZ} ranges over 31 kHz to 330 kHz across the 2 A to 8 A load envelope, so the converter dynamics that the lump is meant to cover are themselves operating-point dependent.
- **The channels are asymmetric by construction.** They run from different input voltages (9 V to 12 V fuel-cell side versus 7.4 V to 8.4 V battery side), so their duty cycles, RHP zeros and crossovers differ even with identical components. Matching the compensator resistor across the two channels was a deliberate hardware change made *so that* a single shared τ_r would be defensible.
- **Measurement noise scales as $1/I_{tot}$.** One ADC code is 8.06 mA, so the share measurement quantises at 0.4 % at 2 A but 1.6 % at 0.5 A, requiring high-frequency roll-off of the actuator signal to avoid DAC chatter.

A controller tuned at the nominal point cannot be assumed to survive this set, and in this case it does not. Evaluated on the discrete closed loop over the parameter corners of Table 3, the legacy $K_p = K_i = 1$ PI reaches a peak sensitivity of $\|S\|_\infty = 26.9$ (28.6 dB), whereas the robust design developed below peaks at $\|S\|_\infty = 1.87$ (5.4 dB). A sensitivity peak of 26.9 corresponds to a disk margin so small that the loop is, for engineering purposes, conditionally stable at the corners — and the corners here are not exotic: they are a cold board, a derated capacitor, and two converters from different reels. This factor-of-14 gap in worst-case sensitivity peak, not any improvement in nominal step response, is the justification for a robust synthesis.

4.3. $\mathcal{H}_\infty$ Synthesis and Youla-H Gain Adjustment

4.3.1. Mixed-sensitivity problem and weight selection

The controller is synthesised by mixed-sensitivity $\mathcal{H}_\infty$ design on the stacked $S/KS/T$ problem, followed by the Youla-H post-synthesis gain adjustment of Tan et al. that enforces exact unity DC complementary sensitivity. The delay is represented by a second-order Padé approximant, so the nominal augmented plant has four states.

Table 4. Synthesis weights and their design intent.

Weight	Acts on	Form / values	Intent
W_p	S	strictly proper $Ma/(s+a)$, $M = 10^4$, unity crossing $40\,\mathrm{rad\,s^{-1}}$	integral-act
W_d	T	`makeweight(0.5, 250, 40)`	force T roll
W_u	Y	`makeweight(0.3, 600, 20)`	cap actuato

W_p is deliberately chosen *strictly* proper. Since $S \to 1$ at high frequency needs no penalty, this costs nothing in design terms but makes the augmented plant satisfy $D_{11} = 0$ exactly, so the regular DGKF problem applies and the estimation Riccati equation degenerates analytically ($D_{21} = 1 \Rightarrow Q_y = 0$, $A - B_1C_2$ stable $\Rightarrow Y = 0$), leaving a single Riccati solve and the central controller $\dot{\hat{x}} = (A - B_1C_2 + B_2F)\hat{x} + B_1e$, $u = F\hat{x}$. Because no commercial robust-control toolbox was available on the development machine, the Riccati solutions were computed from the Hamiltonian matrix by ordered real Schur decomposition with γ-iteration by bisection, in a purpose-written implementation whose primitives are unit-tested against reference solvers and whose every synthesised controller passes an *a-posteriori* gate: closed-loop stability plus $\|T_{zw}\|_\infty \le \gamma$ evaluated by an independent Hamiltonian-bisection norm. A derivation error therefore cannot silently yield an accepted controller.

4.3.2. Synthesis result and the $T(0) = 1$ step

Table 5. Synthesis outcome (nominal plant).

Quantity	Value
γ_{opt}	0.6532
Controller built at	$\gamma = 0.6859$ (5 % back-off), $\|T_{zw}\|_\infty = 0.6841$ ✓
$\mathcal{H}_\infty$ controller order	7 (4 plant + 3 weight states)
$T_H(0)$ before adjustment	0.999 964 23
$T_{YH}(0)$ after adjustment	1.000 000 000 000
S/T crossover	$109.9\,\mathrm{rad\,s^{-1}}$
$\|S\|_\infty$, $\|T\|_\infty$, $\|Y\|_\infty$	1.25, 1.00, 1.01
Nominal M_2 / phase margin / delay margin	0.7978 / 73.9° / 11.74 ms

That $\gamma_{opt} < 1$ with margin means every weighted specification is met, but the $\mathcal{H}_\infty$ solution exhibits the classical DC deficiency: $T_H(0) = 0.999\,964\,23$, short of unity by 0.0036 %. For this application that residual is not cosmetic. It leaves a finite DC loop gain, so the static mismatch disturbance d of Eq. (14) — which can be 0.17 in share at 2 A — is only attenuated, not rejected, and a ramping EMS reference accrues a persistent bias.

The Youla-H step applies the scalar gain correction $Y_{YH} = Y_H/T_H(0)$ and recovers the controller as $G_{C,YH} = Y_{YH}(1 - G_P Y_{YH})^{-1}$. This 0.0036 % gain shift places an exact integrator in the controller: the near-origin pole of $G_{C,YH}$ appears at $|p| = 5.6 \times 10^{-9}$ and is snapped to a true integrator by the spectral split of the following subsection. The outcome is $T(0) = 1$ to machine precision, hence exact DC disturbance rejection and zero ramp-following bias, obtained *without* re-tuning the weights and without disturbing the $\mathcal{H}_\infty$ robustness the synthesis just bought.

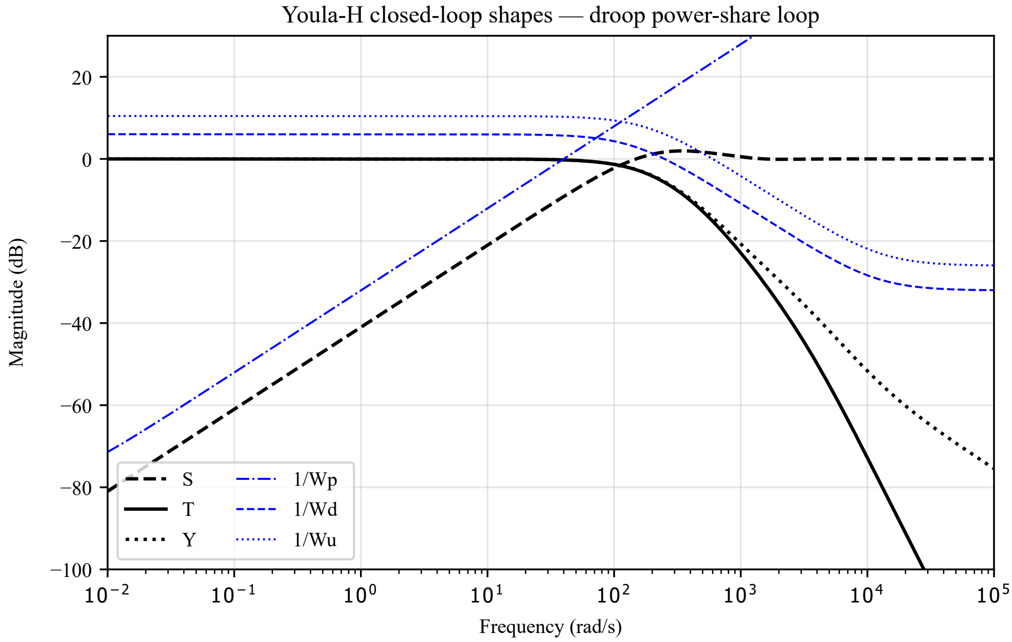

Figure 3. Achieved sensitivity S, complementary sensitivity T and control sensitivity Y of the Youla-H design, plotted against the inverse weights $1/W_p$, $1/W_d$, $1/W_u$. All three lie inside their templates, consistent with $\gamma_{opt} = 0.6532$.

4.3.3. Order reduction, discretisation, and firmware realisation

The controller is decomposed as $G_{C,YH}(s) = k_I/s + R(s)$ with $k_I = 111.93$ — consistent with the observed crossover, since $|L| = 1$ where $k_I/\omega \approx 1$ — and the 21-state stable remainder $R(s)$ is reduced by balanced truncation to **three states**. The Hankel singular values (14.8, 0.035, 0.0018, 5×10^{-5}, …) justify the cut, and the reduced controller's frequency response deviates from the full one by at most 0.8 %.

Tustin discretisation at $T_s = 1$ ms yields three second-order sections plus a trapezoidal integrator. The embedded implementation evaluates

$$u_k = r_0 + R(z)e_k + I(z)e_k, \qquad r_0 = 0.5, \qquad r_k = \mathrm{sat}_{[r_{min}, r_{max}]}(u_k), \tag{16}$$

as a cascade of Direct-Form-II-transposed biquads in single precision, with the output clamped to $[0.15, 0.85]$. Anti-windup is by **back-calculation on the output clamp**: when u_k would leave the interval, the integrator state absorbs exactly the excess, so the command sits on the rail and resumes moving on the tick the error reverses. The biquad states are deliberately *not* clamped — $R(z)$ is stable and cannot wind up, so clamping it would only introduce a spurious nonlinearity. The measured share (only — never the set-point) passes through the 200 Hz one-pole prefilter that is part of the design plant as τ_f, so EMS steps are not smoothed by a sensor filter. A wrapper gates the update to exactly one execution per T_s and holds the output between ticks; that zero-order hold is itself accounted for in T_d.

Two implementation notes are worth recording. First, one discrete section carries a pole at $z = 0.999\,996$ — the W_p weight pole reappearing nearly cancelled in the controller. Its distance from the unit circle, 4×10^{-6}, is about 30× single- precision epsilon at unity, so the section is representable, and its Hankel singular value (0.035) shows it genuinely contributes rather than being truncation noise. Second, the loop freezes (states held, no integration) when $I_{tot} \approx 0$, matching the validity limit of Eq. (14), where α is undefined.

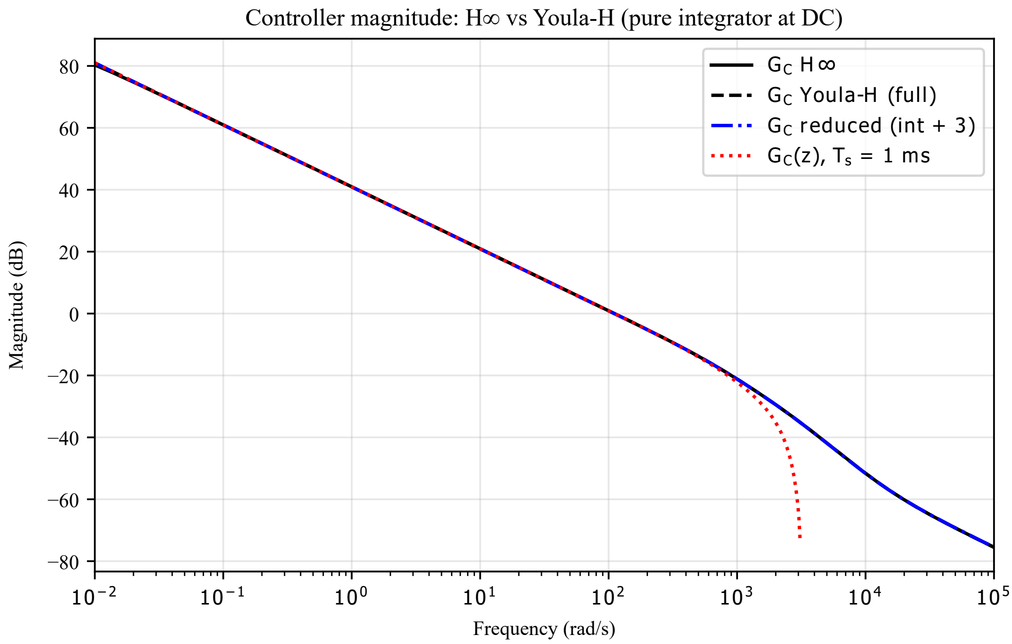

Figure 4. Controller frequency response at each stage of the realisation: the raw $\mathcal{H}_\infty$ controller, the Youla-H adjusted controller, the reduced form, and the discretised implementation. The exact integrator introduced by the Youla-H step is visible as the diverging low-frequency branch that the raw $\mathcal{H}_\infty$ controller lacks.

4.4. Validation

4.4.1. Corner sweeps and time-domain behaviour

The design was checked over the full uncertainty family of Table 3: $K \in \{0.55 \ldots 1.45\}$, $T_d \in \{0.5, 1, 2\}$ ms, $\tau_r \in \{20, 300\}$ µs, $\tau_f \in \{0, 0.8\}$ ms, i.e. **60 corners**. All 60 continuous-time closed loops are stable, with worst-case $M_2 = 0.5905$ occurring at $(K, T_d, \tau_r, \tau_f) = (1.45, 2\,\text{ms}, 300\,\mu\text{s}, 0.8\,\text{ms})$; the corresponding discrete sweep (zero-order-hold plant against the *implemented* $G_C(z)$) has all closed-loop poles inside the unit circle.

Table 6. Validation summary. All entries generated by the synthesis pipeline.

Check	Result
Continuous 60-corner sweep	all stable, worst $M_2 = 0.5905$
Discrete 60-corner sweep	all $\lvert z \rvert < 1$
Nominal margins	GM 18.5 dB, PM 73.9° at 110 rad s^{-1}, delay margin 11.74 ms
Step 0.5 → 0.7 (discrete)	2 % settle 22 ms, 0 % overshoot, SS error 8×10^{-9}
Ramp 0.05 s^{-1}	tracking error 4.47×10^{-4}, no accumulating bias
Anti-windup (0.5 s into the rail)	leaves rail ≤ 3 ticks after error reversal, recovery < 150 ms
Worst-corner discrete $\lVert S \rVert_\infty$	1.867 (legacy PI: 26.889)

The delay margin of 11.74 ms deserves emphasis: it is roughly 6× the worst *modelled* loop delay of 2 ms. Since the timing parameters are pre-calibration estimates, this margin — not any single nominal number — is what makes the design safe to commit to firmware ahead of bench identification.

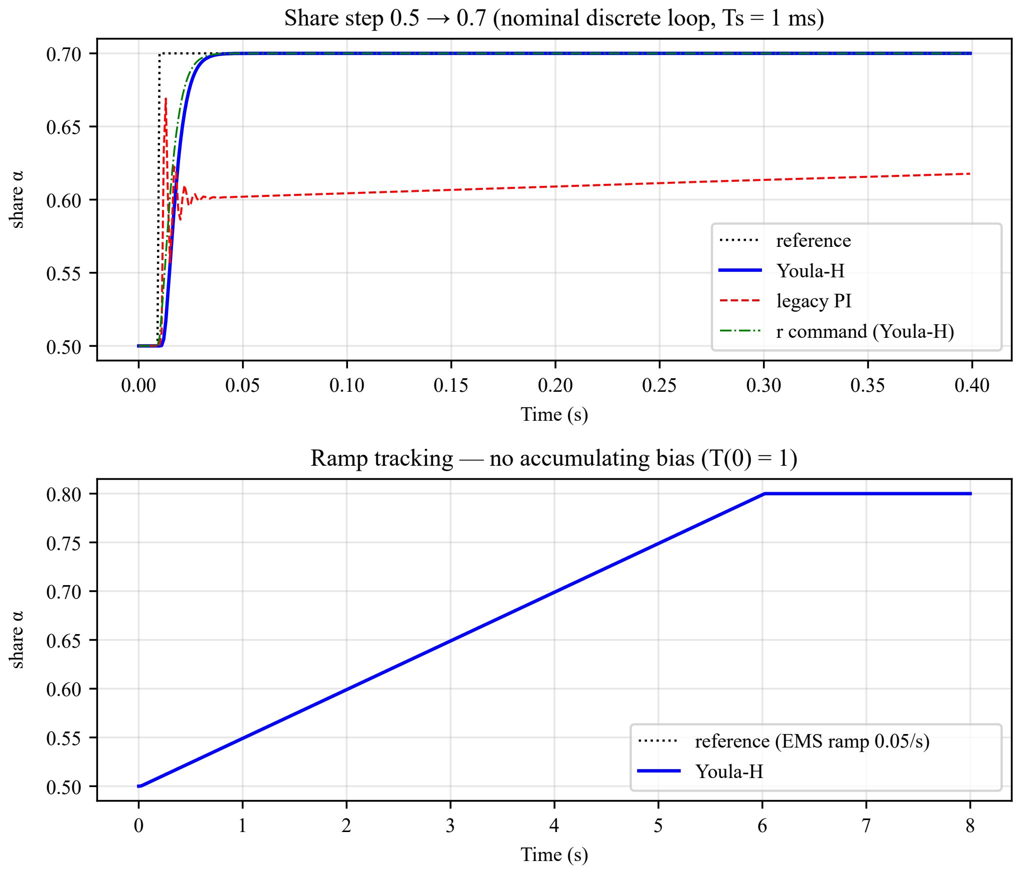

Figure 5. Discrete-time closed-loop response of the implemented controller: share step $0.5 \to 0.7$ (left) and ramp reference at $0.05\,\mathrm{s}^{-1}$ (right). The ramp panel is the practical expression of $T(0) = 1$: the tracking error decays to 4.5×10^{-4} instead of settling to a constant offset.

PLACEHOLDER — plot from `step_response_nominal.csv` / `step_response_mismatch.csv`

Figure 6. Open-loop verification of the static share law, Eq. (14), from the independent numerical model: commanded ratio r, achieved share α, and bus voltage. With matched channels ($\Delta V_0 = 0$) the share tracks r instantaneously; the companion mismatch case ($\alpha = 0.48$ at $r = 0.40$) shows the open-loop offset that only integral action can remove.

4.4.2. Full-order cross-validation

The design plant (15) rests on lumping two converter voltage loops into one uncertain lag. To test that reduction rather than merely argue it, a complete small-signal model of the share plant was constructed *independently*: both channels' transconductance-amplifier compensators with the exact COMP-node impedance $Z_{comp} = R_{EA}\|(R_C + 1/sC_C)\|1/sC_P$, Norton-equivalent power stages carrying the right-half-plane and ESR zeros, the three-resistor feedback node with the bilinear droop injection linearised at (g_0, I_0), current-sense dynamics, bus coupling, and the same digital layer — **11 states** per operating point, against two states plus a delay in the design model.

Structural gate checks confirm the construction before any comparison is drawn: the Norton form reduces to the datasheet power-stage expression to machine precision (4.5×10^{-16}); the DC share gain emerging from the full linearisation is 0.9933, short of unity only by the finite error-amplifier DC gain; and the per-channel voltage-loop crossovers reproduce the independent datasheet arithmetic to within 1 % to 3 % (e.g. FC 16.33 kHz computed versus 16.84 kHz predicted).

Table 7. Full-order versus simplified plant, and closed-loop equivalence with the shipped controller.

Check	**Full-order model**	**Simplified mod**
Nominal in-band deviation ($\omega \leq 1100\,\mathrm{rad\,s^{-1}}$)	3.86 %	—
Envelope deviation, 432 operating points	5.78 % worst, 4.30 % median	—
Discrete closed-loop stability	**432/432** points	60/60 corners
Worst discrete $\|S\|_\infty$	1.240	1.867
Step 0.5 $\rightarrow$ 0.7 overlay	max divergence 0.0008 share	—
2 % settle	25 ms	22 ms
Ramp error at 3 s	4.30×10^{-4}	4.47×10^{-4}

The 432-point envelope spans $\{V_{in,FC}\ 9/12\} \times \{V_{in,BT}\ 7.4/8.4\} \times \{I_{tot}\ 1/2/4\,\mathrm{A}\} \times \{r_0\ 0.3/0.5/0.7\} \times \{C_O\ 30/66\,\mu\mathrm{F}\} \times \{C_{bus}\ 30/500\,\mu\mathrm{F}\} \times \{R_{EA}\ 1/10/100\,\mathrm{M\Omega}\}$; the amplifier output resistance is swept rather than assumed because the datasheet does not specify it, converting a hidden assumption into a verified non-sensitivity.

Two results matter. First, every full-order response lies within 5.78 % of *some* member of the simplified corner family, so the uncertainty set used for synthesis was a genuine superset of the true dynamics. Second, the worst full-order closed-loop sensitivity peak (1.240) is *lower* than the simplified family's worst (1.867) — the expected sign, confirming that the corner family was pessimistic in the safe direction rather than accidentally optimistic in some unmodelled one.

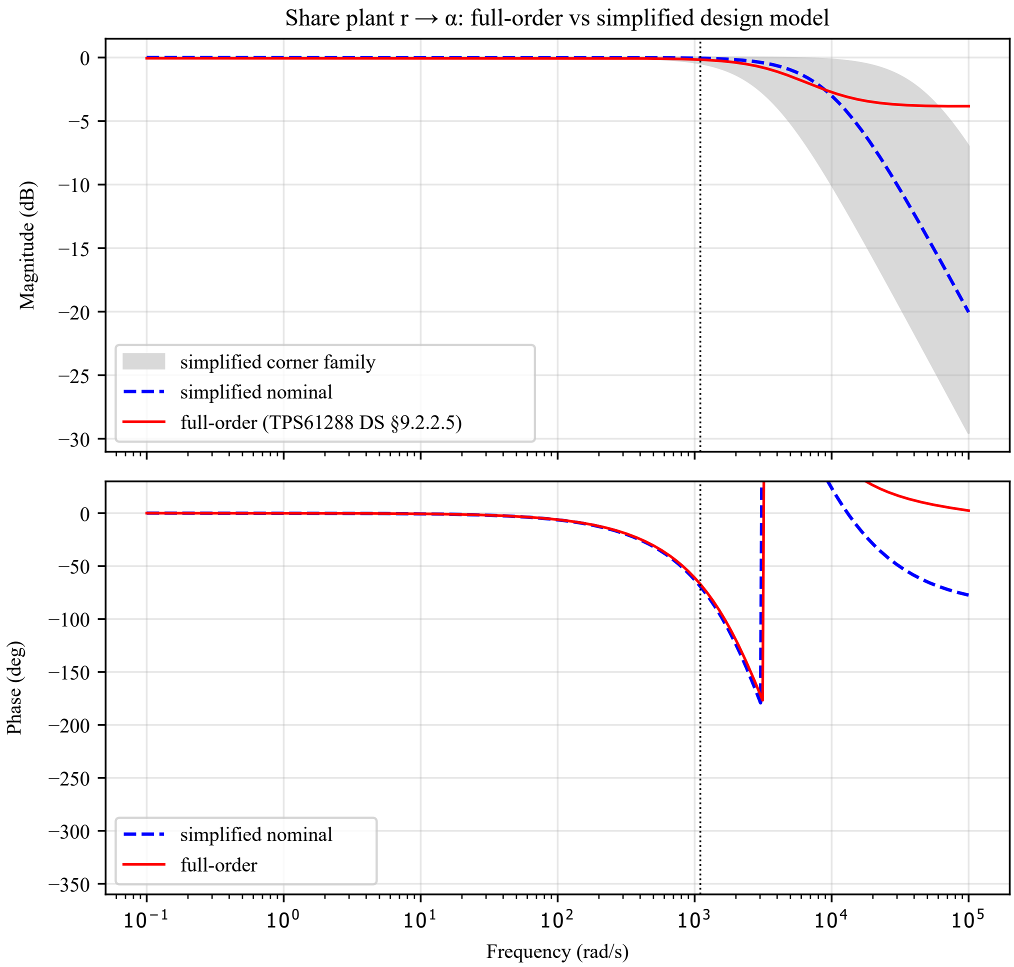

Figure 7. Nominal open-loop frequency response of the 11-state full-order model overlaid on the two-state-plus-delay design plant. The traces are visually indistinguishable through the design band ($\omega \leq 1100\,\mathrm{rad\,s^{-1}}$, max deviation 3.86 %) and separate only beyond $\approx 10^4\,\mathrm{rad\,s^{-1}}$, where the share-loop gain is far below unity.

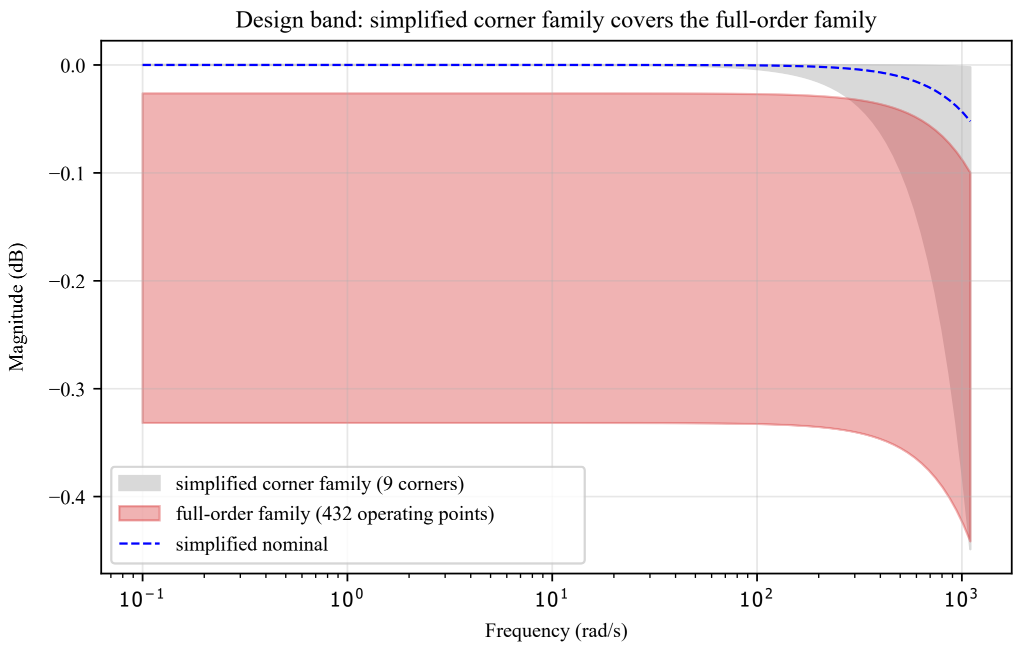

Figure 8. Envelope study: 432 full-order operating points against the simplified corner family. The corner family bounds the full-order family with margin, which is the condition for the synthesis uncertainty set to be valid.

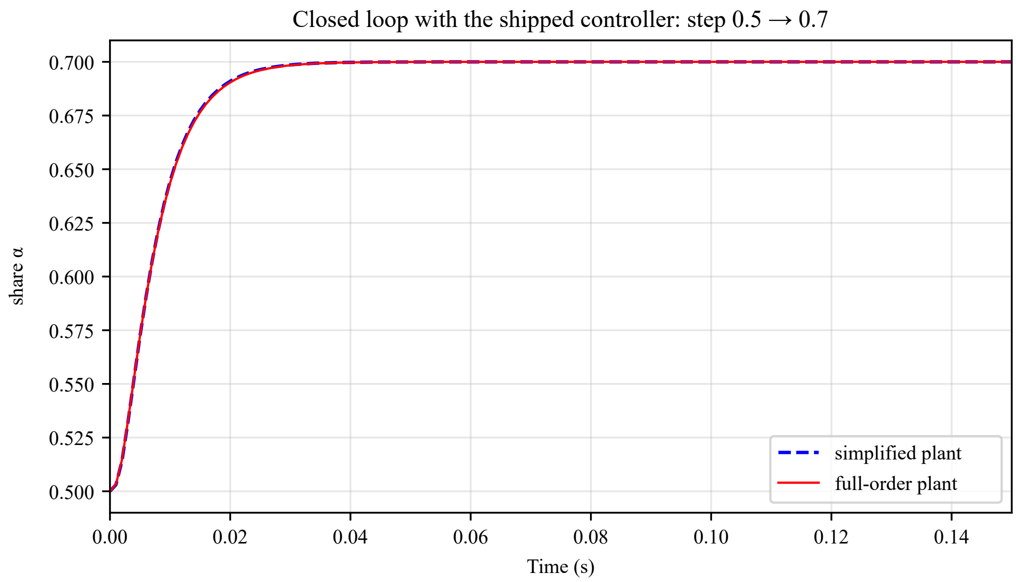

Figure 9. Closed-loop step response of the shipped controller on the full-order plant overlaid on the design prediction. Maximum divergence is 0.0008 share.

4.4.3. Independent toolchain cross-check

Because the synthesis was performed in a purpose-written solver, the entire result was re-derived in a commercial robust-control toolbox using the same weights, and the *shipped* coefficients — parsed directly from the firmware header rather than re-synthesised — were re-run through the validation battery in the firmware-accurate loop topology.

Table 8. Independent toolbox cross-check of the shipped controller.

Quantity	Independent toolbox	Design pipeline
γ_{opt}	0.6553	0.6532 (+0.3 %)
$T_H(0)$	0.999 963 78	0.999 964 23
k_I	110.429	111.930 (+1.36 %)
Crossover	109.4 rad s^{-1}	109.9 rad s^{-1}
Nominal M_2	0.796	0.798
Corner sweep	30/30 stable, worst $\|S\|_\infty = 1.768$	60/60 stable, worst 1.867
Step / DC gain	25 ms settle, DC gain 1.000 000 001	22 ms settle

The two independent syntheses agree on γ_{opt} to 0.3 % and on the nominal margins to within measurement resolution; the 9.5 % frequency-response deviation between the shipped and re-synthesised controllers is attributable to the 21 $\to$ 3 order reduction plus two independent γ-iterations, and the meaningful comparison — closed-loop behaviour and corner stability — agrees closely.

4.5. Limitations

Three qualifications bound the claims above. First, the plant timing parameters (T_d, τ_r, C_{bus}, ΔV_0 and the droop scale k_d) are pre-calibration estimates; the corner sweep is the hedge, and the design is deliberately delay-tolerant (11.7 ms margin against 2 ms worst modelled). A bench identification procedure — open-loop r-sweep to fit ΔV_0 and confirm unity slope, plus a step test on the current-sense outputs to extract τ_r and the true T_d — is defined, after which the coefficients are regenerated through the same automated pipeline. Second, the plant gain is assumed positive over the operating set; below about 1 A total current with worst-case mismatch the gain envelope widens beyond the design set, so the supervisor must avoid commanding extreme shares at near-zero load (the loop is in any case frozen as $I_{tot} \to 0$). Third, the analysis is small-signal and continuous-conduction; pulse-frequency-mode operation at light load is unmodelled, though even a 1 ms effective lag would cost only 6.3° at the share crossover, less than the phase already budgeted by the 2 ms delay corner.

5. Balancer Board Design and Droop Implementation Hardware

5.1. Overall Architecture

The balancer board (revision 20260622) implements a two-source, bidirectional power-pathing architecture for a hybrid fuel-cell/battery scale-model vehicle. Two independently regulated sources are combined onto a common DC bus: a hydrogen fuel cell and a 2S lithium battery pack, each boosted to the nominal bus voltage by an independent TPS61288 synchronous boost converter (FC channel and BT channel). The two boost outputs are not simply diode-ORed onto the bus; each is gated onto the bus through a dedicated RT1987 ideal-diode power-path switch, so the firmware – not the analog hardware alone – decides when each source is allowed to contribute current.

Downstream of the bus, power flows through a further ideal-diode switch to the VESC motor controller and the drive motor (`MOT_PWR_ENABLE`). The motor is also the vehicle's regenerative braking source: during deceleration the VESC can source current back onto the DC bus. This current is intentionally *not* routed back through the boost converters – doing so would require running a TPS61288 in reverse outside its intended operating envelope and risks the back-feed hazard discussed in Section 5.3. Instead, regenerated energy is routed through a dedicated `REGEN_ENABLE` path to the on-board battery charger.

Battery charging is handled by two cooperating circuits with very different bandwidths:

- A **TL431/BSP170P shunt (chopper) clamp** is the *fast*, board-local protection element. It is a fixed analog circuit (TL431 reference, BSP170P shunt MOSFET, dump resistor) with *no firmware involvement* – it exists to absorb the leading edge of a regen transient before any slower control loop can react, and to bound bus overvoltage independent of MCU state.
- A **Silvertel Ag105 MPPT module** is the *secondary*, slow harvester. It receives either fuel-cell bus power (`FC_CHARGE_ENABLE`) or regen power (`REGEN_ENABLE` + `MOT_PWR_ENABLE`) and trickles it into the 2S pack under its own perturb-and-observe MPPT loop. Firmware controls it coarsely, through an active-low `MPPT_DISABLE` GPIO and a two-register I^2C configuration write (voltage/current profile selection), not through a fast current-command interface.

A Teensy 4.1 (ARM Cortex-M7, 600 MHz) is the sole controller on the board. It executes the droop power-share control loop, the motor current loop (talking to the VESC over UART), the sequencing state machine for the ideal-diode switches, and the balancer/charger supervisory logic, and exposes telemetry/command framing over UDP to a Raspberry Pi bridge. A BQ29200 provides hardware-independent overvoltage protection (OVP) on the individual battery cells; its balance-enable pin is hardwired inactive, and only its OVP disable (`CBAL_DISABLE`) is under firmware control.

5.2. Droop Implementation Hardware Chain

Each boost channel implements current-mode droop by perturbing its own voltage-feedback divider in proportion to measured output current, so that as a channel's current rises, its regulated bus voltage target sags slightly – the classical mechanism by which multiple sources sharing one bus settle to a current split without a fast communication link between them. On this board, the droop *coefficient* is not fixed in resistor value; it is a *digitally commandable gain*, which is the key hardware enabler for a closed-loop, MCU-mediated power-share controller rather than a purely passive droop network. The signal chain per channel is:

1. **Current sense – INA253A1IPWR.** A bidirectional, integrated-shunt current-sense amplifier sits at each boost channel's output, converting channel current to a ground-referenced analog voltage. The board specified the A3 variant ($0.4\,\mathrm{V\,A^{-1}}$) but the A1 variant ($0.1\,\mathrm{V\,A^{-1}}$) was fitted by procurement error; firmware compensates with the corrected gain ($K_{\mathrm{sns}} = 0.1\,\mathrm{V\,A^{-1}}$).
2. **Buffer/scale – OPA197.** A rail-to-rail op-amp buffers and scales the sense voltage before it is (a) read back by the Teensy ADC for telemetry and share-error computation, and (b) used as the injection driver downstream of the MDAC.
3. **Digitally-controlled attenuation – AD5443 multiplying DAC.** A 12-bit MDAC (SPI, one per channel, independent chip-selects `CS_MDAC_FC / CS_MDAC_BT`) sits in the injection path and acts as a firmware-programmable attenuation element. This is what turns "droop coefficient" from a soldered resistor value into a run-time variable the power-share controller can command directly.
4. **Feedback injection network.** The MDAC output is summed into each boost converter's own voltage-feedback divider through a dedicated injection resistor, so the converter's regulation setpoint is pulled down in proportion to (measured current) $\times$ (MDAC code), closing the droop loop at the analog level while the MCU sets only the gain.

Because the injection network taps into the converter's own compensated feedback loop, the achievable droop range is bounded by the converter's loop dynamics, not just by the MDAC's 12-bit resolution – this is the link between the board-level hardware described here and the plant model used in the controller synthesis (Section 4). The MCU-side resource cost of this scheme is deliberately modest: one shared SPI bus, two chip selects, two ADC channels per source for sense readback, and no dedicated timer/PWM hardware – the droop actuation is fully absorbed by the analog injection network, leaving the MCU free to run the share controller as a periodic software loop rather than a hardware peripheral.

5.3. Power-Pathing: Ideal-Diode Switches and Sequencing

Six RT1987 ideal-diode power-path switches, each independently controlled by a Teensy GPIO, gate every power path on the board:

`FC_BUS_ENABLE`	FC boost $\rightarrow$ bus
`BT_BUS_ENABLE`	BT boost $\rightarrow$ bus
`MOT_PWR_ENABLE`	bus $\rightarrow$ VESC/motor node
`REGEN_ENABLE`	regen node $\rightarrow$ charger
`FC_CHARGE_ENABLE`	bus (FC) $\rightarrow$ charger
`BT_SEQUENCE_ENABLE`	battery pack sequencing

This is a deliberate departure from a simpler "both boosts always paralleled on the bus" design. Two hazards drive the sequencing constraints:

- **Back-feed / disabled-converter stress.** A converter that is disabled but still electrically connected to an energized bus can see reverse current or overshoot stress it was not designed for. Consequently, whenever the firmware brings a path up or down, the switch ordering is chosen so a regen or charge path is never left pointed at a converter that is not actively driving that node.
- **Hot-plug onto a discharged bulk capacitance.** Closing an ideal-diode switch onto a large discharged capacitor (the bus and, behind `MOT_PWR_ENABLE`, a 470 µF node plus the VESC's own input capacitance) presents a near-short to the source side for the duration of the switch's soft-start ramp. If that ramp cannot supply the inrush within the switch's short-circuit protection window, the switch enters a burst-retry limit cycle, and the resulting repeated high-current transients have – empirically, on this board – destroyed boost converters (Section 6). This is why bus and motor-node bring-up is staged (switches first, allowed to settle, *then* boosts; the motor node is pre-charged before `MOT_PWR_ENABLE` is closed) rather than done in one step.

The explicit rule set (mirrored in the firmware's power-path sequencing guard) is: `BT_SEQUENCE_ENABLE` initializes off and is latched on once during bring-up; `FC_CHARGE_ENABLE` requires `BT_BUS_ENABLE` and `REGEN_ENABLE` to both be low before it is asserted (charger input path and regen path are mutually exclusive); and `MOT_PWR_ENABLE` is only asserted onto a pre-charged, sufficiently energized bus.

As a defense-in-depth measure independent of firmware correctness, every enable net carries a 10 kΩ pull-down to ground, so that during MCU reset or any high-impedance boot window, every switch defaults open (off) rather than floating into an undefined, possibly-closed state.

5.4. Major Components

Table 9. Major active components on the balancer board (rev. 20260622) and their role in the droop / power-pathing architecture.

Component	Part	Role
FC / BT boost regulator	TI TPS61288	Per-source boost to the DC bus, hosts droop injection
Ideal-diode power-path switch ($\times 6$)	Richtek RT1987	Digitally gated, reverse-blocking power path
Current-sense amplifier	TI INA253A1IPWR	Per-channel forward-current sense ($K_{sns} = 0.1\,\mathrm{V/A}$)
Droop MDAC	Analog Devices AD5443	12-bit, SPI-programmable droop-injection gain
MDAC output buffer	TI OPA197	Rail-to-rail output driving injection network
Battery charger / MPPT	Silvertel Ag105	Secondary harvester, I^{2}C config + `MPPT_DISABLE` GPIC
Fast regen clamp	TI TL431 + Infineon BSP170P	Passive, board-local overvoltage / regen shunt
Cell OVP	TI BQ29200	2S cell overvoltage protection, `CBAL_DISABLE` controlle
MCU	NXP i.MX RT1062 (Teensy 4.1)	Droop, sequencing, motor-loop, and telemetry control
Motor controller	VESC (external)	UART motor current control, regen source

5.5. Figures

PLACEHOLDER — system block diagram; see TODO comment in source for required content

Figure 10. Block diagram of the balancer board power and control architecture. *[Placeholder – diagram to be produced from the schematic.]*

PLACEHOLDER — power-path topology diagram; see TODO comment in source

Figure 11. Power-path topology showing the six RT1987 ideal-diode switches, their sequencing constraints, and the hazard points that motivate them. *[Placeholder – to be produced from the schematic.]*

PLACEHOLDER — annotated board photo; see TODO comment in source

Figure 12. Assembled balancer board (rev. 20260622), annotated. *[Placeholder – photo pending final bench assembly.]*

6. Bring-Up Failures, Modeling-Driven Diagnosis, and Fixes

The power-sharing controller developed in Section 4.3 was designed against a small-signal plant that presumes a functioning pair of boost converters delivering current into a common DC bus. Reaching that operating point on hardware proved substantially harder than reaching it in simulation: five TPS61288 boost stages were destroyed over a six-week bring-up campaign, none of them while the droop loop was closed. This section reports that campaign because its outcome is methodologically relevant rather than merely anecdotal. Two of the three root causes were identified only after building quantitative models—a Gerber-derived parasitic-inductance extraction and a soft-start charge-transfer timing model—and both models resolved questions that no amount of additional bench iteration had resolved in four prior attempts. We also report the hypotheses that were wrong, and the evidence that superseded them, because the discarded hypotheses were each individually plausible and are each individually the standard textbook answer.

6.1. Chronology of the Converter Failures

All five failures shared an identical post-mortem signature: the converter's input, switch, and output nodes fused to ground ($V_{\mathrm{BT}} \to \mathrm{GND}$ or $V_{\mathrm{FC}} \to \mathrm{GND}$ short), consistent with destruction of the synchronous-rectifier FET. Table 10 summarizes the conditions.

Table 10. Converter failure events during bring-up. Deaths 1–4 are battery-channel; Death 5 is fuel-cell-channel.

#	Source	Action	Outcome
1	Two 9 V batteries	Enable BT→bus ideal-diode switch	BT boost destroyed (original,
2	Bench supply, 120 mA limit	Firmware bring-up: switches, then boosts	Supply collapsed into curren
3	Bench supply, ≥5 A	Manual bring-up command	5 A to 7 A with collapsed rail,
4	Bench supply, 8.2 V/200 mA	Known-good FC part transplanted to BT pad	Immediate current-limit peg;
5	FC from stiff bench supply, VESC attached	Close motor-node switch at full bus	FC boost destroyed

Four hypotheses were advanced and subsequently rejected. *(i) Capacitive inrush.* The obvious explanation for a converter dying at the instant it is connected to a bus is inrush into bulk capacitance. This was refuted topologically: the 470 µF bulk capacitor sits on the motor node *behind* a separate ideal-diode switch, not on the shared bus, and that switch was open in every one of Deaths 1–4. The bus itself carries only the 30 µF to 40 µF of ceramics distributed across the ideal-diode controllers. *(ii) Source collapse and restart cycling.* Because the logic rail is derived from the battery input through a linear regulator, a soft source sags under converter load, browns out the microcontroller, and provokes a re-enable cycle—a genuine failure mode observed on the bench. It was rejected as *the* cause because Death 3 occurred from a stiff $\geq$5 A supply that never collapsed, and Death 2 occurred at 120 mA, where the deliverable input energy is orders of magnitude below the destruction energy. *(iii) Reverse conduction through the disabled converter.* An early design note asserted that a disabled boost presents a body-diode path through which regenerative energy back-feeds the synchronous rectifier. Reading the RT1987 ideal-diode datasheet retired this: the part uses back-to-back integrated FETs and provides full isolation when disabled. The claim survived in the design documentation for several weeks after it was falsified, and propagated into firmware sequencing comments—a documentation-hygiene failure worth reporting. *(iv) Compensation-network mismatch.* The battery channel's compensation resistor differed from the fuel-cell channel's; matching them was the intervention in Death 4, which still destroyed the part.

Death 4 is the decisive experiment. A converter that had been *proven good* regulating on the fuel-cell pad was transplanted to the battery pad, verified to regulate standalone, and then destroyed on its first bus connection. Combined with three non-destructive control experiments—driving the bus node directly from a supply with the converter removed (clean, $\approx$0 A); driving the bus from the fuel-cell converter (clean); and OR-ing a live bus against a supply through the battery-side ideal diode (clean)—the fault was localized to the *battery channel's physical copper*, with the schematic confirmed identical part-for-part. That conclusion, however, was reached by elimination, and elimination does not produce a mechanism. Producing one required the modeling described next.

6.2. Parasitic-Inductance Extraction from Fabrication Gerbers

The surviving hypothesis was that the battery channel's output-capacitor commutation loop—the "hot loop" carrying the discontinuous current pulse when the synchronous rectifier commutates—enclosed enough area to ring the switch node past the device's 20 V absolute-maximum rating under bus-drive di/dt, with only 0.5 V to 1.5 V of nominal headroom available. Physical measurement on the assembled board found the battery channel's output ceramics placed 240 mil from the converter output pin against 40 mil on the fuel-cell channel. A placement ratio, however, is not an inductance ratio: all power nets on this board are polygon pours, so conductor *length* is a poor proxy and an earlier trace-length-based estimate was later withdrawn as an artifact of a parser that excluded the pours.

We therefore built a dedicated extraction tool (`tools/inductance/`) that computes loop inductance directly from the fabrication data, with no manual geometry re-entry:

1. **Gerber and Excellon parsing.** A self-contained RS-274X and drill parser converts the manufacturing files into copper polygons and hole locations, so the extracted geometry is the geometry that was actually fabricated.
2. **Net isolation by flood fill.** Gerbers carry no net names; the forward net (a converter-output pour) and the return net (the ground pour on the opposite layer) are selected by coordinate and flood-filled at a fine isolation pitch, then resampled to a coarser meshing pitch. Plane-clearance slots are morphologically bridged so the return plane remains a single connected body at every pitch, with a severed-plane detector that flags meshing artifacts.
3. **PEEC solution.** The meshed forward and return conductors are emitted as a uniform-mesh FastHenry deck at their true stackup separation, with an explicit port and an explicit loop closure at the load, and solved for $L(f) = \mathrm{Im}\{Z_{\mathrm{port}}(f)\}/2\pi f$ over DC to 1 MHz.
4. **Independent bound.** A closed-form microstrip estimate is computed from the same mesh as a sanity check on the numerical result.

Because a PEEC result is only as trustworthy as its discretization, the tool solves two orthogonal sweeps rather than a cross product: a *mesh-convergence* axis (all mesh pitches at one designated frequency) and a *frequency* axis (all frequencies at the converged pitch), sharing the designated point. Solutions are persisted per-point to an append-as-you-go store, making multi-hour runs interruptible and idempotent.

The result is given in Table 11. At the converged mesh, the battery-channel output loop extracts to 3.48 nH at DC falling to 3.15 nH at 1 MHz (skin effect), against 1.54 nH and 1.46 nH for the fuel-cell channel—a ratio of 2.2× at 1 MHz. The two channels' extracted loop *lengths* (9.69 mm versus 4.74 mm) and effective widths (5.91 mm versus 12.47 mm) show the asymmetry is both a longer and a narrower return path.

Table 11. Extracted output-loop inductance, converged mesh.

Channel	Mesh (mm)	L (DC)	L (1 MHz)	Loop len.	Eff. width
Fuel cell	0.3	1.538 nH	1.456 nH	4.74 mm	12.47 mm
Battery	0.2	3.480 nH	3.149 nH	9.69 mm	5.91 mm

The indicated remedy is a board-level one—firmware cannot reduce a loop inductance. Ceramic capacitors of 10 µF and 0.1 µF were bodged directly at the battery converter's output pins, collapsing the 240 mil loop to approximately the fuel-cell channel's geometry. The validation was deliberately single-variable: an otherwise exact repetition of the Death 4 configuration, differing only in the added capacitors. Where every unmodified attempt had destroyed a converter, the modified channel survived *four consecutive* bring-ups, regulating the bus normally (Figures 13–15).

Two limitations qualify this result. First, all validation captures were taken at 1× probe bandwidth (≈10 MHz, 50 MSa/s); the hypothesized 100 MHz to 200 MHz hot-loop ring is invisible at that bandwidth. *Survival is the evidence, not the waveform*, and a high-bandwidth switch-node margin measurement remains owed. Second, the extraction models one user-selected loop at a time and is not a full-board three-dimensional solve; it supports a comparative claim between two nominally identical channels far better than it supports an absolute ring-amplitude prediction.

6.3. Soft-Start and Protection Timing in the Ideal-Diode Path

The hot-loop fix held for the battery channel, but a fifth converter—this time on the fuel-cell side, with good layout—was destroyed while closing the motor-node switch onto an attached motor controller. This failure has a different character and is, in our view, the more generalizable one, because it arises from the *interaction of two independently designed control loops* rather than from a layout defect.

The bus-to-motor path is gated by an RT1987 ideal-diode controller whose inrush behavior is set by an open-loop soft-start ramp: a fixed external capacitance ($C_{SS} = 5.6\,\mathrm{nF}$) sets a turn-on time of order 1.1 ms, during which a start-up short-circuit protection (SCP) foldback current limit is armed with a 250 µs continuous-clamp timer and a 64 ms retry interval. The ramp time is a designed constant; it does not adapt to the load it is charging. Downstream sits the motor node—470 µF of bulk capacitance plus the motor controller's own (unmeasured) input capacitance, a total measured near 590 µF on the bare board. The charge demand of that node over the fixed ramp, $I \approx 0.8\,C\,V/t_{\mathrm{ON}}$, lands in the same amperes-class range as the applicable foldback limit. The ideal diode's protection therefore engages or nearly engages on connection, and each clamp-and-release cycle presents the upstream boost converters with a large, abrupt load step followed by an equally abrupt load dump.

Neither loop is misdesigned in isolation. The ideal diode's soft start is a correct open-loop inrush limiter for a bounded load capacitance; the boost's voltage loop is a correctly compensated regulator with a 4 kHz to 19 kHz crossover. The hazard lives in their coupling: the load-dump transient at the end of a clamp interval occurs on a timescale comparable to the boost's loop latency, so the boost error amplifier is left wound up delivering current into a suddenly unloaded output. The resulting overshoot is then *peak-held* by the ideal diode itself, which cannot sink current—the switch behaves as a peak detector on the bus, and the rail parks above nominal until leakage discharges it. Measured parks of 0.65 V to 0.72 V (centre-to-centre) above the regulation point were observed, sufficient to trip the firmware bus overvoltage fault on roughly 80 % of bring-up attempts once the bus nominal was retuned downward. At the 15 A-class currents of a full-bus motor-node connection, the same event scaled past the converter's absolute-maximum rating and destroyed it—but only from a stiff source, since a battery source sags into undervoltage lockout before lethal current flows. This source dependence retroactively explains the otherwise puzzling pattern across all five failures.

Two remedies follow directly, and are complementary. The firmware remedy is *sequencing*: never close the motor-node switch at full bus voltage. The bring-up now raises the motor-node switch together with the bus switches *before* the boost converters ramp, so the converters' own soft start charges the entire downstream stack from a low initial voltage in one coordinated ramp; the node is then kept energized through idle and run states, with the motor held stopped by a zero current command rather than by removing power. A guard predicate faults rather than permitting a hot connection if the bus and motor-node measurements indicate the node is discharged. This trades away hardware motor isolation in idle—a deliberate and documented posture change. The hardware remedy is to size the ideal diode's soft-start capacitance to the load it actually charges: increasing C_{SS} from 5.6 nF to 100 nF extends the ramp to tens of milliseconds and reduces the connect demand to a few hundred milliamperes, restoring roughly an order of magnitude of margin. A subsequent datasheet reconciliation also revealed that the firmware's inter-phase settle delay had been sized from the soft-start time while omitting the controller's 8 ms turn-on *delay*, so the intended coordinated ramp was not in fact executing; the current recommendation is to replace the fixed delay with an ADC-confirmed pre-charge gate and timeout rather than any tuned constant.

6.4. Relationship to the Droop Architecture

It is important to state precisely which of these hazards the droop power-sharing architecture created, and which it merely inherited. Table 12 gives the attribution.

Table 12. Attribution of each bring-up hazard to architecture-specific versus generic causes.

Hazard	Attribution	Basis
Output hot-loop overshoot (Deaths 1–4)	Generic	A single-channel boost driving any capacitive bus exhibits it. The droop injection network was inactive (DACs unwritten) in every failure; the asymmetry is fabrication layout, present before any control code ran.
Motor-node hot-plug load dump (Death 5)	Generic multi-converter	Arises from an ideal-diode soft-start versus a downstream capacitive node; identical in a single-source system with the same power-path switching.
Ideal-diode peak-hold overvoltage parking	Generic	A consequence of non-sinking sources behind unidirectional switches; independent of how current is shared between them.
Source-collapse restart cycling	Generic (bench)	An artifact of deriving logic power from a converter input rail on a soft supply.
Droop-injection gain mapping error	**Droop-specific**	A firmware mapping omitted the feedback-injection attenuation, saturating both DACs and presenting the share loop a zero-gain plant. Non-destructive, but it invalidated all share-loop bench data taken before it was found (Section 3.2.3).
Compensation-resistor lever	**Coupled**	The obvious mitigation for load-dump overshoot—raising the boost crossover—is constrained because the compensation resistor sets the response time constant of the share-loop plant model, so a unilateral change would invalidate the synthesized controller.

The summary judgement is that the destructive failures were *multi-converter power-path bring-up hazards*, not droop-control hazards: they occurred with the sharing loop open, with the modulating DACs at their reset state, and on a single-channel basis. The droop architecture's genuine contribution to the campaign was a non-destructive but consequential one—a gain-mapping error that silently opened the share loop—and a constraint on the remedy space, in that the compensation network is shared between the local voltage loop and the outer sharing plant and cannot be retuned for transient margin without re-synthesizing the controller. That coupling is, we would argue, the transferable control-level lesson: in a droop-shared converter pair, the per-converter compensation network is no longer a purely local design variable.

6.5. Figures

PLACEHOLDER — source: `references/scope_captures/1-VOUT.jpg`

Figure 13. Battery-converter output during a validated bring-up with hot-loop capacitors fitted: body-diode pre-charge, one aborted soft-start attempt with automatic retry, then clean regulation. The retry is a benign source-sag undervoltage event, and is the same class of event that destroyed converters before the fix.

PLACEHOLDER — source: `references/scope_captures/5-Vout yellow-Vmot blue-Ibat purple.jpg`

Figure 14. Three-channel capture through a bring-up: converter output, bus voltage, and current-sense output. The wide current hump is the downstream node charging through a soft start that completes; the subsequent output park is the peak-held overshoot.

PLACEHOLDER — source: `references/scope_captures/3-SW.jpg`, `4-SW zoomed in.jpg`

Figure 15. Switch-node envelope during soft start. The initial discrete pulses are normal pulse-frequency-modulation skipping, not instability. Note the bandwidth limitation discussed in Section 6.2.

PLACEHOLDER — generate from `tools/inductance` frequency-sweep store

Figure 16. Extracted loop inductance versus frequency for both converter output loops, with the mesh-convergence axis inset and the closed-form microstrip bound shown for reference.

PLACEHOLDER — to be drawn

Figure 17. Campaign timeline and fault tree. Upper band: chronological failure and intervention events. Lower band: hypothesis tree, with each branch annotated by the experiment that refuted or confirmed it.

7. Toward a Combined MIMO Power–Drive Controller

The architecture presented so far treats the power-share loop (droop ratio r, Section 4) and the drive loop (motor current command from the velocity controller) as decoupled SISO problems. In reality the two loops are coupled through the bus: a drive transient changes the total load current, which both channels' droop resistances convert into a share disturbance, while a share transient modulates the bus voltage seen by the motor drive.

7.1. Planned Comparison

- Augment the share plant with the drive dynamics (VESC current loop approximated as a first-order lag + delay; motor/vehicle mechanical mode) to form a 2×2 MIMO plant: inputs (droop ratio command, motor current command), outputs (measured share, wheel speed).
- Quantify the off-diagonal coupling (RGA / σ-plots) to establish whether the decentralized design is actually justified or merely convenient.
- Synthesize a MIMO H_∞ controller on the same weight philosophy as Section 4 and compare against the decentralized pair on: worst-corner $\|S\|_\infty$, drive-transient share excursion, and regen-event bus excursion.
- Assess implementation cost on the Teensy 4.1 (state count vs. the current three-biquad realization at 1 kHz).

TODO(figure): 2×2 MIMO block diagram — droop/share loop and drive loop with cross-coupling paths highlighted

Figure 18. Planned MIMO structure combining the power-share and drive loops. *(Placeholder — analysis not yet performed.)*

8. Conclusions

This paper demonstrated that controllable power sharing between heterogeneous DC sources can be implemented entirely with off-the-shelf regulator ICs by electronically injecting a commanded droop resistance into each converter's feedback node, and that the resulting share loop — despite wide converter tolerance bands and channel asymmetry — can be made robust by H_∞/Youla synthesis against an uncertainty-corner plant family.

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